# Return to Play After Shoulder Instability Surgery in National Collegiate Athletic Association Division I Intercollegiate Football Athletes

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**Background:** Recent attention has focused on the optimal surgical treatment for recurrent shoulder instability in young athletes. Collision athletes are at a higher risk for recurrent instability after surgery.

**Purpose:** To evaluate variables affecting return-to-play (RTP) rates in Division I intercollegiate football athletes after shoulder instability surgery.

**Study Design:** Case series; Level of evidence, 4.

**Methods:** Invitations to participate were made to select sports medicine programs that care for athletes in Division I football conferences (Pac-12 Conference, Southeastern Conference [SEC], Atlantic Coast Conference [ACC]). After gaining institutional review board approval, 7 programs qualified and participated. Data on direction of instability, type of surgery, time to resume participation, and quality and level of play before and after surgery were collected.

**Results:** There were 168 of 177 procedures that were arthroscopic surgery, with a mean 3.3-year follow-up. Overall, 85.4% of players who underwent arthroscopic surgery without concomitant procedures returned to play. Moreover, 15.6% of athletes who returned to play sustained subsequent shoulder injuries, and 10.3% sustained recurrent instability, resulting in reduction/revision surgery. No differences were noted in RTP rates in athletes who underwent anterior labral repair (82.4%), posterior labral repair (92.9%), combined anterior-posterior repair (84.8%; $P = .2945$), or open repair (88.9%; $P = .9362$). Also, 93.3% of starters, 95.4% of utilized players, and 75.7% of rarely used players returned to play. The percentage of games played before the injury was 49.9% and rose to 71.5% after surgery ($P < .0001$). Athletes who played in a higher percentage of games before the injury were more likely to return to play; 91% of athletes who were starters before the injury returned as starters after surgery. Scholarship status significantly correlated with RTP after surgery ($P = .0003$).

**Conclusion:** The majority of surgical interventions were isolated arthroscopic stabilization procedures, with no statistically significant difference in RTP rates when concomitant arthroscopic procedures or open stabilization procedures were performed. Athletes who returned to play often played in a higher percentage of games after surgery than before the injury, and many played at the same or a higher level after surgery.

**Keywords:** shoulder instability; return to sport; intercollegiate football

The glenohumeral joint is the most common joint to sustain a dislocation, both in the general population as well as in contact and collision sports. A prior study has noted a risk of 0.12 per 1000 exposures in contact sports, with the highest rate occurring during the spring collegiate football training period.[15] Kaplan et al[12] noted a 14% rate of shoulder instability in players presenting to the National Football League Combine. Shoulder instability was the second most common injury among these athletes, with at least 4.7% of Combine athletes having undergone prior shoulder stabilization surgery.[3,12]

Open shoulder surgery that addresses labral injuries with repair and capsular imbrication has been recognized as the "gold standard" for the treatment of shoulder instability.[6] Arthroscopic stabilization surgery has been used with greater frequency, essentially supplanting open stabilization surgery as many patients' and surgeons' treatment of choice.[16] However, concern has been raised with regard to the effectiveness of arthroscopic stabilization over time,

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particularly in high-demand collision and contact athletes.[5,19] Some authors advocate continued participation in contact sports as a relative contraindication to performing arthroscopic repair.[9] As a result, the use of bone block augmentation of the anterior glenoid rim has gained more attention in this particularly challenging population. Neyton et al[14] demonstrated that the Latarjet-Patte procedure provided reliable results in rugby players, with only 4% not returning to sport because of their shoulder.

Despite these concerns, arthroscopic shoulder stabilization surgery remains a common mode of treatment for physicians providing care for American football athletes. The purpose of this study was to evaluate National Collegiate Athletic Association (NCAA) Division I collegiate football athletes who sustained at least 1 shoulder instability event that required surgical stabilization as a means of allowing them to "return to sport" at this highly demanding level of competition. In particular, the rate of return and quality of play before and after surgery were analyzed. We hypothesized that a high percentage of stabilization surgery being performed on this patient cohort would be arthroscopic procedures, that this patient population would have a fairly high return-to-play (RTP) rate, and that these athletes would be able to progress in their respective football programs with regard to the percentage of games played and depth chart status.

## METHODS

Requests for participation in this retrospective study of football athletes who had undergone shoulder stabilization surgery were sent to sports medicine programs that directly care for athletes participating in the Division I football Pac-12 Conference, Southeastern Conference (SEC), and Atlantic Coast Conference (ACC). After gaining institutional review board approval, 7 programs were able to participate in the study. Inclusion criteria included all intercollegiate athletes active on their respective team's rosters during the 2004-2013 seasons who sustained at least 1 shoulder instability event that eventually required surgical stabilization treatment. For inclusion in analysis, athletes had to have at least 1 more season of eligibility after surgery and had to have available data for the rest of their collegiate eligibility. Data on direction of instability, type of surgery, time to resume participation, and quality and level of play both before and after surgery were collected for each athlete. All data points for each team were collected by their respective sports medicine program and collated based on each site's athletic training injury tracking and surveillance database. RTP was defined as being granted clearance by the medical and athletic training staff as well as the athletes returning to their respective team roster.

Data were analyzed to determine overall RTP rates based on the type of shoulder stabilization surgery (anterior, posterior, combined) performed and included concomitant procedures (superior labral from anterior to posterior [SLAP] repair, rotator interval closure, etc). The clockface was used to define the site of the labral tear (anterior: 2:30-6:00; posterior: 6:00-11:00). Open surgical interventions were also noted and recorded. The influence of scholarship status on RTP and ability to complete years of eligibility, graduation, and/or going on to compete at a higher level were recorded. Recurrent shoulder injuries of any type resulting in missed practices or games were recorded, and recurrent instability requiring reduction or the need for revision surgery was also documented.

Quality of play both before and after surgery was measured by several aspects. To determine the effect that surgery had on players' ability to return to play, the percentage of games played compared with games eligible to play before the injury and after surgery was determined. Two of 7 teams were not able to collect these particular data points (20 patients), and subset analysis was performed on the athletes from the other 5 teams for this particular data point. The player's roster status (starter, utilized, rarely played) was noted both before the injury and after surgery. "Utilized" was defined as >1 play per game, while rarely played was defined as 0 to 1 plays per game. Statistical analysis was performed using SAS 9.3 software (SAS Institute) chi-square analysis and the Fisher exact test were used for performing comparative analysis.

## RESULTS

Pre hoc power analysis determined that a sample size of approximately 170 was recommended to detect small to medium effect sizes with 80% power based on chi-square testing.[7] One hundred seventy-seven shoulder injuries in 153 athletes from 7 different institutions belonging to the Pac-12 (n = 4), ACC (n = 1), and SEC (n = 2) were identified and met initial inclusion criteria. In cases in which particular data points were not available or obtainable on a particular athlete, subset analysis was performed. The mean follow-up after stabilization surgery was 3.3 years.

One hundred sixty-two was the subset available from the pool of 177 cases that had data for both MRI and preoperative clinical direction of instability. The preoperative clinical diagnosis based on examination and magnetic resonance imaging findings demonstrated that 50% (81/162) were classified as anterior instability, 38% (62/162) were posterior instability, and 12% (19/162) were a combined or multidirectional pattern. One hundred sixty-eight shoulders underwent arthroscopic surgery, and 9 shoulders (8 patients) underwent open bone block surgery (3

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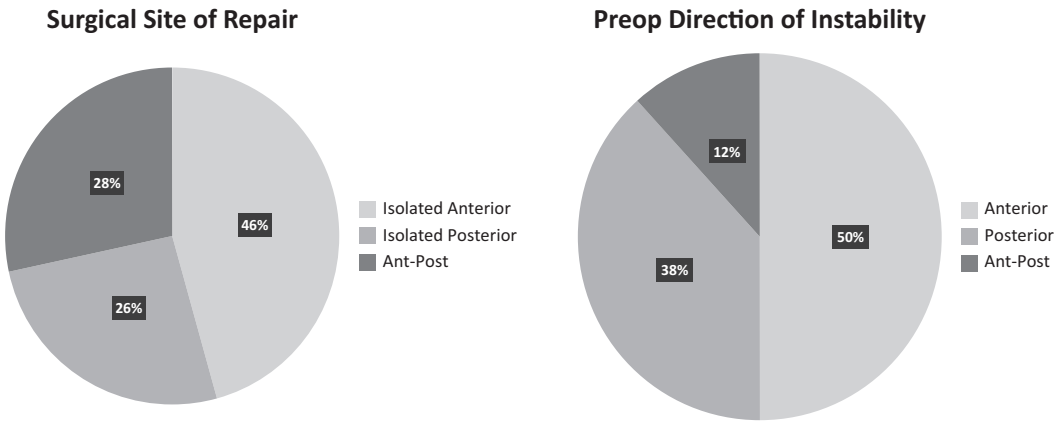


**Figure 1.** Difference in preoperative instability direction versus surgical direction of repair. Ant-Post, anterior-posterior.

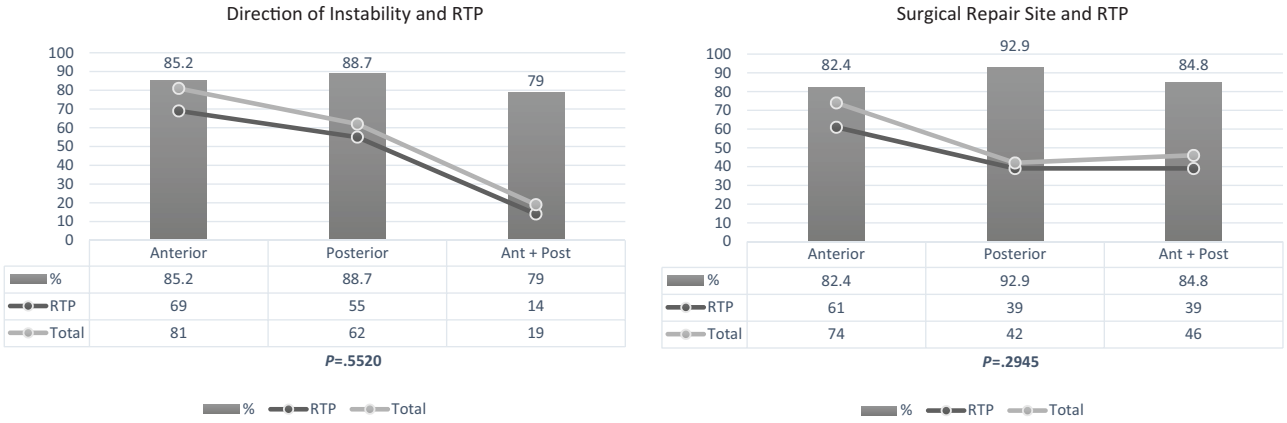


**Figure 2.** Surgery and injury versus return to play (RTP). Ant+Post, anterior-posterior.

Bristow procedures, 5 Latarjet procedures, 1 with open bone block allograft).

At the time of surgical repair in those treated arthroscopically, additional labral lesions were found, necessitating additional work either posterior or anterior in a substantial number of cases in which instability was initially thought to be unidirectional. Of those with the clinical preoperative diagnosis of anterior instability, 7 of 81 (9%) underwent combined anterior-posterior labral repair based on arthroscopic findings, while 20 of 62 (32%) of those diagnosed with posterior instability underwent combined anterior-posterior repair (Figure 1). In other words, athletes who were diagnosed with posterior instability were 3.5 times more likely to undergo combined anterior-posterior repair than athletes diagnosed with anterior instability.

Overall, 85.4% of players who underwent arthroscopic surgery without concomitant procedures returned to play. Moreover, 82.4% of players who underwent anterior labral repair, 92.9% of those who underwent posterior labral repair, and 84.8% who underwent combined anterior-posterior repair returned to play (Figure 2); 87% of those who underwent an additional procedure in addition to arthroscopic labral repair returned to play.

Eight of 9 shoulders in 8 athletes that underwent open anterior stabilization surgery returned to play. One of these athletes who underwent a Latarjet procedure returned to play for a time but quit before his eligibility was completed. Two of the Latarjet procedures and 1 of the Bristow procedures were performed for revision surgery after recurrent instability events after an arthroscopic stabilization procedure. These were in athletes who had successfully returned to play for a period of time but after a reinjury required repeat surgery. All other bony operations were primary procedures surgically indicated by physician or athlete choice and an overall assessment of abnormalities. Subset analysis of 163 cases available for comparison revealed there was no significant difference between RTP rates when comparing open stabilization (n = 9), isolated arthroscopic labral repair (n = 123), and arthroscopic labral repair with combined procedures (n = 31) ($P$ = .9362).

Categorized by depth chart position, 93.3% of starters, 95.4% of utilized players, and 75.7% of rarely used players returned to play (Figure 3). The percentage of games played by athletes before the injury was 49.9% and rose to 71.5% after surgery ($P < .0001$) (Figure 4).

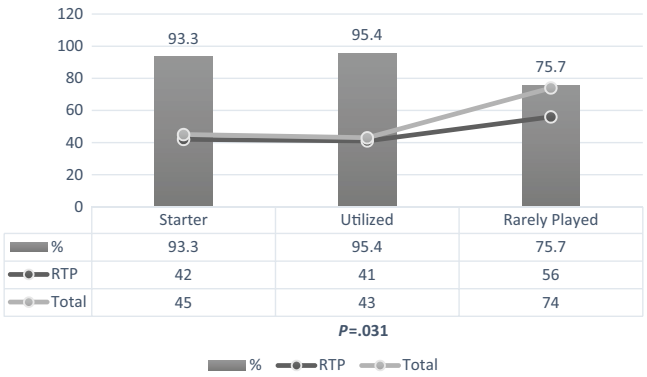


**Figure 3.** Player status before injury and subsequent return to sport. RTP, return to play.

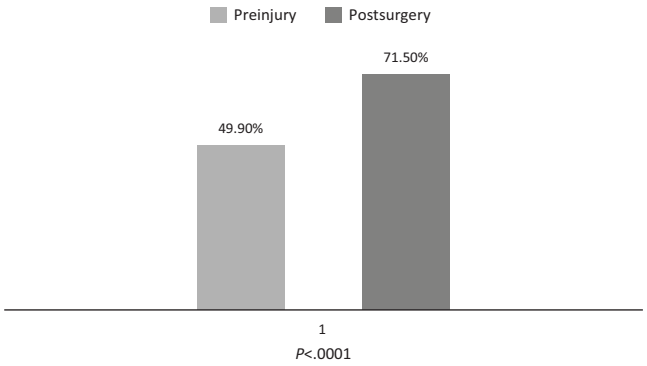


**Figure 4.** Percentage of games played before and after surgery.

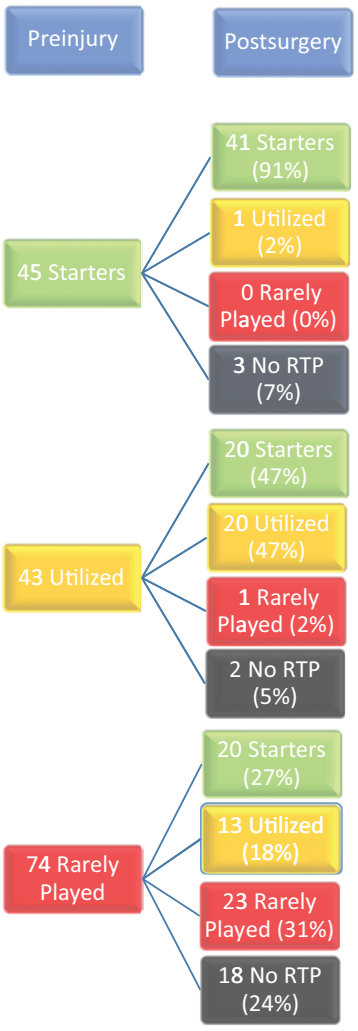


**Figure 5.** Players' status before injury versus after surgery and return to play (RTP).

Of the 45 athletes identified as starters before their injury, 91% continued as starters, while 2% became utilized players after surgery. Of the 43 utilized players before their injury, 47% became starters, 47% remained utilized, 2% rarely played, and 5% did not return to play after surgery. For the 74 players identified as rarely playing before their injury who returned to play, 27% became starters, 18% were utilized athletes, and 31% maintained their rarely playing status (Figure 5). Only 76% of these athletes returned to play. The change in status before and after the injury was statistically significant (McNemar test, $P < .0001$).

Eighty-four percent of athletes who returned to play completed their college eligibility or went on to play at a higher level, 8.5% attained RTP status but did not complete their eligibility, and 7.5% did not complete their eligibility because they did not return to play ($P < .0001$); 78% of athletes cleared for RTP did not go on to play football after college.

Furthermore, 15.6% of athletes who returned to play had some type of a shoulder injury that resulted in missed practices and/or games; 10.3% of all athletes who returned to play were reported to have subsequent instability that required either manual reduction or revision shoulder surgery. No significant difference was found in the RTP rate of those who did and did not have recurrent injuries ($P$ = .9870) or undergo subsequent surgery or manual reduction ($P$ = .8518) when compared with those who did not experience additional shoulder problems.

Being on scholarship significantly correlated with RTP after surgery; 90.9% of those with a scholarship returned to play, while 65.5% of those not on scholarship returned to play ($P$ = .0003). Graduation rates also correlated with RTP rates. Ten percent of athletes who eventually graduated failed to return to play, while 28% of those who did not graduate failed to return to play ($P$ = .0446).

## DISCUSSION

Participation in collision and contact sports has been identified as a significant risk factor for both initial and recurrent shoulder instability.[3,5,11,12,15,21] In addition, Robinson et al[20] identified young age and male sex as significant risk factors for recurrent shoulder instability, with Randelli et al[18] also noted young age to be the most significant risk factor for

recurrent shoulder instability. American college football therefore is an environment in which young male athletes engage in repetitive, high-energy contact activity, creating a relatively high-risk environment for shoulder instability injuries.[15] Our study looked at a population of athletes performing at the highest level of intercollegiate football competition with presumably some of the highest demands and forces placed on athletes and their shoulders. In this unique patient population, 85.4% of all athletes who required surgery for shoulder instability returned to play. In addition, players who returned did so typically at the same or a higher status on the depth chart and played in approximately 20% more games that they were eligible to play in after surgery than they did before their injury.

This study evaluated anterior versus posterior versus combined shoulder instability and found no significant differences in RTP rates. In regards to anterior stability, much attention has been given to arthroscopic versus open versus bone block procedures. It is interesting to note that during the time frame of the study, the majority of procedures performed were arthroscopic in nature. Recent meta-analyses and systematic reviews have demonstrated no difference in recurrence rates between open and modern arthroscopic procedures using suture anchors.[8,17,18] Harris et al,[8] in their systematic review, reported RTP rates of 87% for the arthroscopic approach and 89% for the open approach. Ide et al[11] noted an 80% RTP rate in contact athletes treated with arthroscopic anterior instability repair. Our study's results are similar to these findings, with an RTP rate of 82.4%. No difference in RTP rates was noted between the open bone block and arthroscopic techniques in our study. This result should be interpreted with caution, however, because of the small number of bone block procedures performed in our study; it is likely that this comparison is underpowered to draw any meaningful conclusions. Neyton et al[14] reported on a 0% recurrent instability rate using the Latarjet coracoid transfer procedure in rugby union players with a 65% RTP rate. In comparing studies on rugby to American football, however, it needs to be noted that the typical mechanism of injuries is different, with rugby union players presenting with a greater rate of glenoid bone fractures or bone loss than do American football players.[4,10] To emphasize this point, Larrain et al[13] performed arthroscopic stabilization using suture anchors in rugby players who had no bony defects or poor tissue quality. This resulted in an RTP rate of 100% in the acute repair group and 84% in their recurrent instability group. These results mirror the findings in our study in which American football players, typically treated at the end of the season (nonacute), tended to suffer more from soft tissue injuries than discrete glenoid bone defects. This may explain, in part, the much higher rate of arthroscopic surgery over open and bone block procedures.

In 2 separate studies, Arner et al[1] and Bradley et al[2] demonstrated a high RTP rate in football athletes undergoing arthroscopic posterior labral repair. Our RTP rates are also similar, which is interesting in that our study population includes only Division I intercollegiate football players while many of the athletes included in Bradley et al's[2] study played at the high school level but also included athletes playing at the intercollegiate and professional levels.

RTP used as an endpoint or for determining a good outcome needs to be interpreted with some degree of caution. To qualify RTP rates, our study evaluated the quality of return by looking at both the percentage of games played and the change in utilization of the athletes in their respective football program. Our findings demonstrated an approximate 20% increase in the percentage of games played after surgery compared with before the injury, which was a statistically significant finding. Additionally, players who were starters or utilized before their injury almost always returned at the same or a higher level of utilization. For players who rarely played before their injury, a substantial number of players improved to be utilized or to be starters. Both of these measures demonstrate that not only did players return to play at a relatively high rate but also did so in a way that allowed them to continue to improve and grow as players in their respective football program. In addition, the RTP rate was significantly higher in the preinjury starters and utilized players, with 93.3% and 95.4% returning to play. For players who played rarely before their injury, only 75.7% returned to play. Our findings suggest that in the higher performing athletes, a very high RTP rate can be anticipated, but for players struggling to find a foothold in their program, the chance of RTP is less certain. This finding was also supported by scholarship status, with 90.9% on scholarship returning to play but only 65.5% of those with nonscholarship status returning to play. Reasons for this may be multifactorial to include players not wanting to risk reinjuries without the benefit of a secured scholarship or coaches discouraging players from returning because of a low chance or need for the players to contribute to their respective team's program.

Twenty-two percent of players in this study went on to play football after their college eligibility expired. As expected, 100% of those players who did go on to play professionally did indeed return to play after their shoulder injury in college. Based on data from Kaplan et al,[12] however, prior shoulder surgery can affect the career longevity of National Football League players. In addition, our study demonstrated a statistically significant correlation with RTP and graduation status, with inability to return to play negatively correlating with graduating from college.

This study's retrospective design is a limitation. As a result, the data may be subject to recall or exclusion bias. In addition, 2 of the 7 schools were unable to determine the percentage of games played before the injury versus after surgery. Therefore, only 142 of the players were able to be used for this aspect of the study, which may affect results. However, statistical significance was found when comparing preinjury rates of games played for those who returned to play (49.9%) versus those who did not (19.6%) ($P$ = .0151). Similar statistical significance was seen when comparing the percentage of games played before and after surgery for those who returned to play (Figure 4). Therefore, power was not a question in this case. Patients with instability symptoms who did not require reduction for a dislocation or chose to undergo revision surgery were not recorded in this database. Along these lines, the reasons for those who did not return to play may not be specifically or solely because of complications related to shoulder

instability. In the present study, 15.6% of athletes who returned to play had recurrent shoulder injuries, leading to missed practices and/or games after their RTP, but might not have been specifically limited to recurrent instability symptoms, and therefore, this finding needs to be interpreted with some caution. In addition, the results of this study should not be used for determining "recurrent instability" after surgery, as "RTP," progression in the depth chart, and percentage of games played do not specifically measure recurrent instability. Along these lines, the reasons for those who did not return to play may not be specifically or solely because of complications related to shoulder instability. Finally, this study does not have a control arm for making comparisons as to the positive or negative effects that surgery had on returning athletes' participation in games and depth chart position compared to those who did not sustain a shoulder injury. The strengths of this study, however, include the number of athletes captured belonging to a defined, isolated, and highly challenged patient population exposed to some of the highest risk factors for recurrence and the quality measures used to determine RTP.

## CONCLUSION

The RTP rate in high-level intercollegiate football players undergoing shoulder stabilization surgery was 85.4%. Posterior labral repair, anterior labral repair, and combined labral repair demonstrated no statistical difference in RTP rates. The majority of surgical interventions were isolated arthroscopic stabilization procedures and demonstrated no statistically significant difference in RTP rates when concomitant arthroscopic procedures or open stabilization procedures were performed. Finally, athletes who returned to play often did so in a higher percentage of games after surgery than they did before their injury, and many players were utilized at the same or a higher level by their teams after surgery. Athletes who started, were used frequently in games, and with a scholarship returned to play more frequently than rarely used and/or nonscholarship players. Athletes who returned to play often did so at or above their preinjury utilization status.

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